

Model2

March 24, 2026

```
[2]: !pip install datasets transformers sentence-transformers
```

```
Collecting datasets
  Downloading datasets-4.8.3-py3-none-any.whl.metadata (19 kB)
Collecting transformers
  Using cached transformers-5.3.0-py3-none-any.whl.metadata (32 kB)
Collecting sentence-transformers
  Using cached sentence_transformers-5.3.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: filelock in /opt/conda/lib/python3.13/site-packages (from datasets) (3.24.3)
Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.13/site-packages (from datasets) (2.3.5)
Requirement already satisfied: pyarrow>=21.0.0 in /opt/conda/lib/python3.13/site-packages (from datasets) (22.0.0)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in /opt/conda/lib/python3.13/site-packages (from datasets) (0.4.0)
Requirement already satisfied: pandas in /opt/conda/lib/python3.13/site-packages (from datasets) (2.3.3)
Requirement already satisfied: requests>=2.32.2 in /opt/conda/lib/python3.13/site-packages (from datasets) (2.32.5)
Requirement already satisfied: httpx<1.0.0 in /opt/conda/lib/python3.13/site-packages (from datasets) (0.28.1)
Requirement already satisfied: tqdm>=4.66.3 in /opt/conda/lib/python3.13/site-packages (from datasets) (4.67.1)
Collecting xxhash (from datasets)
  Using cached xxhash-3.6.0-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (13 kB)
Collecting multiprocessing<0.70.20 (from datasets)
  Using cached multiprocessing-0.70.19-py313-none-any.whl.metadata (7.5 kB)
Requirement already satisfied: fsspec<=2026.2.0,>=2023.1.0 in /opt/conda/lib/python3.13/site-packages (from multiprocessing<0.70.20->datasets) (2026.1.0)
Collecting huggingface-hub<2.0,>=0.25.0 (from datasets)
  Downloading huggingface_hub-1.7.2-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: packaging in /opt/conda/lib/python3.13/site-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (25.0)
Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.13/site-
```

packages (from datasets) (6.0.3)
 Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in
 /opt/conda/lib/python3.13/site-packages (from
 fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (3.13.3)
 Requirement already satisfied: anyio in /opt/conda/lib/python3.13/site-packages
 (from httpx<1.0.0->datasets) (4.12.1)
 Requirement already satisfied: certifi in /opt/conda/lib/python3.13/site-
 packages (from httpx<1.0.0->datasets) (2026.1.4)
 Requirement already satisfied: httpcore==1.* in /opt/conda/lib/python3.13/site-
 packages (from httpx<1.0.0->datasets) (1.0.9)
 Requirement already satisfied: idna in /opt/conda/lib/python3.13/site-packages
 (from httpx<1.0.0->datasets) (3.11)
 Requirement already satisfied: h11>=0.16 in /opt/conda/lib/python3.13/site-
 packages (from httpcore==1.*->httpx<1.0.0->datasets) (0.16.0)
 Collecting hf-xet<2.0.0,>=1.4.2 (from huggingface-hub<2.0,>=0.25.0->datasets)
 Using cached
 hf_xet-1.4.2-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata
 (4.9 kB)
 Collecting typer (from huggingface-hub<2.0,>=0.25.0->datasets)
 Using cached typer-0.24.1-py3-none-any.whl.metadata (16 kB)
 Requirement already satisfied: typing-extensions>=4.1.0 in
 /opt/conda/lib/python3.13/site-packages (from huggingface-
 hub<2.0,>=0.25.0->datasets) (4.15.0)
 Collecting dill<0.4.2,>=0.3.0 (from datasets)
 Using cached dill-0.4.1-py3-none-any.whl.metadata (10 kB)
 Collecting regex!=2019.12.17 (from transformers)
 Using cached regex-2026.2.28-cp313-cp313-
 manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
 (40 kB)
 Collecting tokenizers<=0.23.0,>=0.22.0 (from transformers)
 Using cached tokenizers-0.22.2-cp39-abi3-
 manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)
 Collecting safetensors>=0.4.3 (from transformers)
 Using cached safetensors-0.7.0-cp38-abi3-
 manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
 Requirement already satisfied: torch>=1.11.0 in /opt/conda/lib/python3.13/site-
 packages (from sentence-transformers) (2.10.0)
 Requirement already satisfied: scikit-learn in /opt/conda/lib/python3.13/site-
 packages (from sentence-transformers) (1.8.0)
 Requirement already satisfied: scipy in /opt/conda/lib/python3.13/site-packages
 (from sentence-transformers) (1.16.3)
 Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
 /opt/conda/lib/python3.13/site-packages (from
 aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (2.6.1)
 Requirement already satisfied: aiosignal>=1.4.0 in
 /opt/conda/lib/python3.13/site-packages (from
 aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.4.0)
 Requirement already satisfied: attrs>=17.3.0 in /opt/conda/lib/python3.13/site-

```

packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets)
(25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/opt/conda/lib/python3.13/site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/opt/conda/lib/python3.13/site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
/opt/conda/lib/python3.13/site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (0.4.1)
Requirement already satisfied: yarll<2.0,>=1.17.0 in
/opt/conda/lib/python3.13/site-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets)
(1.22.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/opt/conda/lib/python3.13/site-packages (from requests>=2.32.2->datasets)
(3.4.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/opt/conda/lib/python3.13/site-packages (from requests>=2.32.2->datasets)
(2.6.3)
Requirement already satisfied: setuptools in /opt/conda/lib/python3.13/site-
packages (from torch>=1.11.0->sentence-transformers) (80.9.0)
Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.13/site-
packages (from torch>=1.11.0->sentence-transformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-
transformers) (3.6.1)
Requirement already satisfied: jinja2 in /opt/conda/lib/python3.13/site-packages
(from torch>=1.11.0->sentence-transformers) (3.1.6)
Requirement already satisfied: cuda-bindings==12.9.4 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-
transformers) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-
transformers) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-
transformers) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-
transformers) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-
transformers) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-

```

transformers) (12.8.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (11.3.3.83)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (11.7.3.90)

Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.5.8.93)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (2.27.5)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (3.4.5)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.90)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.93)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (1.13.1.3)

Requirement already satisfied: triton==3.6.0 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (3.6.0)

Requirement already satisfied: cuda-pathfinder~=1.1 in /opt/conda/lib/python3.13/site-packages (from cuda-bindings==12.9.4->torch>=1.11.0->sentence-transformers) (1.4.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/lib/python3.13/site-packages (from sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.13/site-packages (from jinja2->torch>=1.11.0->sentence-transformers) (3.0.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.13/site-packages (from pandas->datasets) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.13/site-packages (from pandas->datasets) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /opt/conda/lib/python3.13/site-packages (from pandas->datasets) (2025.3)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)

Requirement already satisfied: joblib>=1.3.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn->sentence-transformers) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.2.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn->sentence-transformers) (3.6.0)

Requirement already satisfied: click>=8.2.1 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface-hub<2.0,>=0.25.0->datasets) (8.3.1)

Collecting shellingham>=1.3.0 (from typer->huggingface-hub<2.0,>=0.25.0->datasets)

Using cached shellingham-1.5.4-py2.py3-none-any.whl.metadata (3.5 kB)

Collecting rich>=12.3.0 (from typer->huggingface-hub<2.0,>=0.25.0->datasets)

Using cached rich-14.3.3-py3-none-any.whl.metadata (18 kB)

Collecting annotated-doc>=0.0.2 (from typer->huggingface-hub<2.0,>=0.25.0->datasets)

Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)

Collecting markdown-it-py>=2.2.0 (from rich>=12.3.0->typer->huggingface-hub<2.0,>=0.25.0->datasets)

Using cached markdown_it_py-4.0.0-py3-none-any.whl.metadata (7.3 kB)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/conda/lib/python3.13/site-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=0.25.0->datasets) (2.19.2)

Collecting mdurl~=0.1 (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=0.25.0->datasets)

Using cached mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)

Downloading datasets-4.8.3-py3-none-any.whl (526 kB)

526.8/526.8 kB

7.5 MB/s 0:00:00

Downloading huggingface_hub-1.7.2-py3-none-any.whl (618 kB)

618.0/618.0 kB

9.8 MB/s 0:00:00

Using cached hf_xet-1.4.2-cp37-abi3-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (4.2 MB)

Using cached multiprocessing-0.70.19-py313-none-any.whl (156 kB)

Using cached dill-0.4.1-py3-none-any.whl (120 kB)

Using cached transformers-5.3.0-py3-none-any.whl (10.7 MB)

Using cached tokenizers-0.22.2-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.3 MB)

Using cached sentence_transformers-5.3.0-py3-none-any.whl (512 kB)

Using cached regex-2026.2.28-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (802 kB)

Using cached safetensors-0.7.0-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (507 kB)

Using cached typer-0.24.1-py3-none-any.whl (56 kB)

Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)

```

Using cached rich-14.3.3-py3-none-any.whl (310 kB)
Using cached markdown_it_py-4.0.0-py3-none-any.whl (87 kB)
Using cached mdurl-0.1.2-py3-none-any.whl (10.0 kB)
Using cached shellingham-1.5.4-py2.py3-none-any.whl (9.8 kB)
Using cached xxhash-3.6.0-cp313-cp313-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (193 kB)
Installing collected packages: xxhash, shellingham, safetensors, regex, mdurl,
hf-xet, dill, annotated-doc, multiprocessing, markdown-it-py, rich, typer,
huggingface-hub, tokenizers, datasets, transformers, sentence-transformers
  Attempting uninstall: dill
    Found existing installation: dill 0.4.0
    Uninstalling dill-0.4.0:90m

```

```

6/17 [dill]
  Successfully uninstalled dill-0.4.0

```

```

6/17 [dill]

```

```

17/17

```

```

[sentence-transformers]ence-transformers]
Successfully installed annotated-doc-0.0.4 datasets-4.8.3 dill-0.4.1 hf-
xet-1.4.2 huggingface-hub-1.7.2 markdown-it-py-4.0.0 mdurl-0.1.2
multiprocess-0.70.19 regex-2026.2.28 rich-14.3.3 safetensors-0.7.0 sentence-
transformers-5.3.0 shellingham-1.5.4 tokenizers-0.22.2 transformers-5.3.0
typer-0.24.1 xxhash-3.6.0

```

```

[4]: !pip install -U datasets transformers sentence-transformers accelerate

```

```

Requirement already satisfied: datasets in /opt/conda/lib/python3.13/site-
packages (4.8.3)
Requirement already satisfied: transformers in /opt/conda/lib/python3.13/site-
packages (5.3.0)
Requirement already satisfied: sentence-transformers in
/opt/conda/lib/python3.13/site-packages (5.3.0)
Collecting accelerate
  Using cached accelerate-1.13.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: filelock in /opt/conda/lib/python3.13/site-
packages (from datasets) (3.24.3)
Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.13/site-
packages (from datasets) (2.3.5)
Requirement already satisfied: pyarrow>=21.0.0 in
/opt/conda/lib/python3.13/site-packages (from datasets) (22.0.0)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in
/opt/conda/lib/python3.13/site-packages (from datasets) (0.4.1)
Requirement already satisfied: pandas in /opt/conda/lib/python3.13/site-packages
(from datasets) (2.3.3)
Requirement already satisfied: requests>=2.32.2 in
/opt/conda/lib/python3.13/site-packages (from datasets) (2.32.5)
Requirement already satisfied: httpx<1.0.0 in /opt/conda/lib/python3.13/site-
packages (from datasets) (0.28.1)

```

Requirement already satisfied: tqdm>=4.66.3 in /opt/conda/lib/python3.13/site-packages (from datasets) (4.67.1)

Requirement already satisfied: xxhash in /opt/conda/lib/python3.13/site-packages (from datasets) (3.6.0)

Requirement already satisfied: multiprocessing<0.70.20 in /opt/conda/lib/python3.13/site-packages (from datasets) (0.70.19)

Requirement already satisfied: fsspec<=2026.2.0,>=2023.1.0 in /opt/conda/lib/python3.13/site-packages (from fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (2026.1.0)

Requirement already satisfied: huggingface-hub<2.0,>=0.25.0 in /opt/conda/lib/python3.13/site-packages (from datasets) (1.7.2)

Requirement already satisfied: packaging in /opt/conda/lib/python3.13/site-packages (from datasets) (25.0)

Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.13/site-packages (from datasets) (6.0.3)

Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /opt/conda/lib/python3.13/site-packages (from fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (3.13.3)

Requirement already satisfied: anyio in /opt/conda/lib/python3.13/site-packages (from httpx<1.0.0->datasets) (4.12.1)

Requirement already satisfied: certifi in /opt/conda/lib/python3.13/site-packages (from httpx<1.0.0->datasets) (2026.1.4)

Requirement already satisfied: httpcore==1.* in /opt/conda/lib/python3.13/site-packages (from httpx<1.0.0->datasets) (1.0.9)

Requirement already satisfied: idna in /opt/conda/lib/python3.13/site-packages (from httpx<1.0.0->datasets) (3.11)

Requirement already satisfied: h11>=0.16 in /opt/conda/lib/python3.13/site-packages (from httpcore==1.*->httpx<1.0.0->datasets) (0.16.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /opt/conda/lib/python3.13/site-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (1.4.2)

Requirement already satisfied: typer in /opt/conda/lib/python3.13/site-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (0.24.1)

Requirement already satisfied: typing-extensions>=4.1.0 in /opt/conda/lib/python3.13/site-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (4.15.0)

Requirement already satisfied: regex!=2019.12.17 in /opt/conda/lib/python3.13/site-packages (from transformers) (2026.2.28)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /opt/conda/lib/python3.13/site-packages (from transformers) (0.22.2)

Requirement already satisfied: safetensors>=0.4.3 in /opt/conda/lib/python3.13/site-packages (from transformers) (0.7.0)

Requirement already satisfied: torch>=1.11.0 in /opt/conda/lib/python3.13/site-packages (from sentence-transformers) (2.10.0)

Requirement already satisfied: scikit-learn in /opt/conda/lib/python3.13/site-packages (from sentence-transformers) (1.8.0)

Requirement already satisfied: scipy in /opt/conda/lib/python3.13/site-packages (from sentence-transformers) (1.16.3)

Requirement already satisfied: psutil in /opt/conda/lib/python3.13/site-packages (from accelerate) (7.2.1)

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (2.6.1)

Requirement already satisfied: aiosignal>=1.4.0 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.4.0)

Requirement already satisfied: attrs>=17.3.0 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (25.4.0)

Requirement already satisfied: frozenlist>=1.1.1 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.8.0)

Requirement already satisfied: multidict<7.0,>=4.5 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (6.7.1)

Requirement already satisfied: propcache>=0.2.0 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (0.4.1)

Requirement already satisfied: yarll<2.0,>=1.17.0 in /opt/conda/lib/python3.13/site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2026.2.0,>=2023.1.0->datasets) (1.22.0)

Requirement already satisfied: charset_normalizer<4,>=2 in /opt/conda/lib/python3.13/site-packages (from requests>=2.32.2->datasets) (3.4.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/conda/lib/python3.13/site-packages (from requests>=2.32.2->datasets) (2.6.3)

Requirement already satisfied: setuptools in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (80.9.0)

Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (3.6.1)

Requirement already satisfied: jinja2 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (3.1.6)

Requirement already satisfied: cuda-bindings==12.9.4 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.9.4)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.93)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-

transformers) (12.8.90)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.90)

Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (9.10.2.21)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (11.3.3.83)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (11.7.3.90)

Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.5.8.93)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (2.27.5)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (3.4.5)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.90)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (12.8.93)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (1.13.1.3)

Requirement already satisfied: triton==3.6.0 in /opt/conda/lib/python3.13/site-packages (from torch>=1.11.0->sentence-transformers) (3.6.0)

Requirement already satisfied: cuda-pathfinder~=1.1 in /opt/conda/lib/python3.13/site-packages (from cuda-bindings==12.9.4->torch>=1.11.0->sentence-transformers) (1.4.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/lib/python3.13/site-packages (from sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.13/site-packages (from jinja2->torch>=1.11.0->sentence-transformers) (3.0.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.13/site-packages (from pandas->datasets) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.13/site-packages (from pandas->datasets) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /opt/conda/lib/python3.13/site-packages (from pandas->datasets) (2025.3)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)

Requirement already satisfied: joblib>=1.3.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn->sentence-transformers) (1.5.3)

Requirement already satisfied: threadpoolctl>=3.2.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn->sentence-transformers) (3.6.0)

Requirement already satisfied: click>=8.2.1 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface-hub<2.0,>=0.25.0->datasets) (8.3.1)

Requirement already satisfied: shellingham>=1.3.0 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface-hub<2.0,>=0.25.0->datasets) (1.5.4)

Requirement already satisfied: rich>=12.3.0 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface-hub<2.0,>=0.25.0->datasets) (14.3.3)

Requirement already satisfied: annotated-doc>=0.0.2 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface-hub<2.0,>=0.25.0->datasets) (0.0.4)

Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/conda/lib/python3.13/site-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=0.25.0->datasets) (4.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/conda/lib/python3.13/site-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=0.25.0->datasets) (2.19.2)

Requirement already satisfied: mdurl~=0.1 in /opt/conda/lib/python3.13/site-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=0.25.0->datasets) (0.1.2)

Using cached accelerate-1.13.0-py3-none-any.whl (383 kB)

Installing collected packages: accelerate

Successfully installed accelerate-1.13.0

```
[6]: !pip install -U accelerate
```

Requirement already satisfied: accelerate in /opt/conda/lib/python3.13/site-packages (1.13.0)

Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.13/site-packages (from accelerate) (2.3.5)

Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.13/site-packages (from accelerate) (25.0)

Requirement already satisfied: psutil in /opt/conda/lib/python3.13/site-packages

(from accelerate) (7.2.1)

Requirement already satisfied: pyyaml in /opt/conda/lib/python3.13/site-packages (from accelerate) (6.0.3)

Requirement already satisfied: torch>=2.0.0 in /opt/conda/lib/python3.13/site-packages (from accelerate) (2.10.0)

Requirement already satisfied: huggingface_hub>=0.21.0 in /opt/conda/lib/python3.13/site-packages (from accelerate) (1.7.2)

Requirement already satisfied: safetensors>=0.4.3 in /opt/conda/lib/python3.13/site-packages (from accelerate) (0.7.0)

Requirement already satisfied: filelock>=3.10.0 in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (3.24.3)

Requirement already satisfied: fsspec>=2023.5.0 in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (2026.1.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (1.4.2)

Requirement already satisfied: httpx<1,>=0.23.0 in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (0.28.1)

Requirement already satisfied: tqdm>=4.42.1 in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (4.67.1)

Requirement already satisfied: typer in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (0.24.1)

Requirement already satisfied: typing-extensions>=4.1.0 in /opt/conda/lib/python3.13/site-packages (from huggingface_hub>=0.21.0->accelerate) (4.15.0)

Requirement already satisfied: anyio in /opt/conda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate) (4.12.1)

Requirement already satisfied: certifi in /opt/conda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate) (2026.1.4)

Requirement already satisfied: httpcore==1.* in /opt/conda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate) (1.0.9)

Requirement already satisfied: idna in /opt/conda/lib/python3.13/site-packages (from httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate) (3.11)

Requirement already satisfied: h11>=0.16 in /opt/conda/lib/python3.13/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface_hub>=0.21.0->accelerate) (0.16.0)

Requirement already satisfied: setuptools in /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (80.9.0)

Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (3.6.1)

Requirement already satisfied: jinja2 in /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (3.1.6)

Requirement already satisfied: cuda-bindings==12.9.4 in

/opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (12.9.4)
 Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.8.93)
 Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.8.90)
 Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.8.90)
 Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (9.10.2.21)
 Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.8.4.1)
 Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (11.3.3.83)
 Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (10.3.9.90)
 Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (11.7.3.90)
 Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.5.8.93)
 Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (0.7.1)
 Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (2.27.5)
 Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate) (3.4.5)
 Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.8.90)
 Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (12.8.93)
 Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
 /opt/conda/lib/python3.13/site-packages (from torch>=2.0.0->accelerate)
 (1.13.1.3)
 Requirement already satisfied: triton==3.6.0 in /opt/conda/lib/python3.13/site-
 packages (from torch>=2.0.0->accelerate) (3.6.0)
 Requirement already satisfied: cuda-pathfinder~=1.1 in
 /opt/conda/lib/python3.13/site-packages (from cuda-
 bindings==12.9.4->torch>=2.0.0->accelerate) (1.4.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/lib/python3.13/site-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.13/site-packages (from jinja2->torch>=2.0.0->accelerate) (3.0.3)

Requirement already satisfied: click>=8.2.1 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface_hub>=0.21.0->accelerate) (8.3.1)

Requirement already satisfied: shellingham>=1.3.0 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface_hub>=0.21.0->accelerate) (1.5.4)

Requirement already satisfied: rich>=12.3.0 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface_hub>=0.21.0->accelerate) (14.3.3)

Requirement already satisfied: annotated-doc>=0.0.2 in /opt/conda/lib/python3.13/site-packages (from typer->huggingface_hub>=0.21.0->accelerate) (0.0.4)

Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/conda/lib/python3.13/site-packages (from rich>=12.3.0->typer->huggingface_hub>=0.21.0->accelerate) (4.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/conda/lib/python3.13/site-packages (from rich>=12.3.0->typer->huggingface_hub>=0.21.0->accelerate) (2.19.2)

Requirement already satisfied: mdurl~=0.1 in /opt/conda/lib/python3.13/site-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface_hub>=0.21.0->accelerate) (0.1.2)

```
[8]: import os
import re
import json
import math
import random
import shutil
from dataclasses import dataclass
from typing import Dict, List, Tuple, Optional

import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim

from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForCausalLM
from sentence_transformers import SentenceTransformer

# =====
# Config
```

```

# =====
@dataclass
class Config:
    # Dataset
    train_subset: int = 200
    test_subset: int = 50

    # Models
    llm_name: str = "Qwen/Qwen2.5-1.5B-Instruct"
    # llm_name: str = "Qwen/Qwen2.5-3B-Instruct"
    embed_name: str = "sentence-transformers/all-MiniLM-L6-v2"

    # RL training
    epochs: int = 5
    policy_learning_rate: float = 5e-4
    value_learning_rate: float = 1e-3
    entropy_coef: float = 0.003
    value_loss_coef: float = 0.5
    grad_clip_norm: float = 1.0

    # Reward
    reward_token_penalty: float = 0.0002
    use_soft_reward: bool = True
    soft_reward_floor: float = 0.0

    # Thinking budget range
    min_budget: int = 64
    max_budget: int = 220

    # Final answer generation budget
    answer_max_new_tokens: int = 20

    # Generation
    do_sample: bool = False
    temperature: float = 0.0

    # Exploration warmup
    warmup_epochs: int = 2
    warmup_mix: float = 0.35

    # Hardware
    seed: int = 42
    device: str = "cuda" if torch.cuda.is_available() else "cpu"

    # Logging
    print_every: int = 10

```

```

debug_examples: int = 3

# Cache
cache_dir: str = "./cache_rl_two_stage_v5"
clear_cache_on_start: bool = True

# Checkpoints
best_policy_path: str = "best_policy_v5.pt"
best_value_path: str = "best_value_v5.pt"
final_policy_path: str = "two_stage_continuous_budget_policy_v5_final.pt"
final_value_path: str = "two_stage_value_net_v5_final.pt"

CFG = Config()

# =====
# Utilities
# =====

def set_seed(seed: int) -> None:
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    if torch.cuda.is_available():
        torch.cuda.manual_seed_all(seed)

def ensure_dir(path: str) -> None:
    os.makedirs(path, exist_ok=True)

def reset_cache_dir(cache_dir: str) -> None:
    if os.path.exists(cache_dir):
        shutil.rmtree(cache_dir)
    os.makedirs(cache_dir, exist_ok=True)

def clamp_budget(x: float, min_budget: int, max_budget: int) -> int:
    return int(max(min_budget, min(max_budget, round(x))))

def denormalize_budget(x: float, min_budget: int, max_budget: int) -> float:
    return x * (max_budget - min_budget) + min_budget

def normalize_scalar_list(values: List[float]) -> List[float]:

```

```

    if len(values) == 0:
        return values
    arr = np.array(values, dtype=np.float32)
    mean = float(arr.mean())
    std = float(arr.std()) + 1e-8
    return ((arr - mean) / std).tolist()

def clean_model_output(text: str) -> str:
    if text is None:
        return ""

    cleaned = text

    stop_markers = [
        "\nHuman:",
        "\nAssistant:",
        "\nUser:",
        "\nSystem:",
        "Human:",
        "Assistant:",
        "User:",
        "System:",
        "<|im_end|>",
        "<|endoftext|>",
    ]

    for marker in stop_markers:
        if marker in cleaned:
            cleaned = cleaned.split(marker)[0]

    return cleaned.strip()

# =====
# GSM8K helpers
# =====

def extract_gsm8k_final_answer(answer_text: str) -> str:
    if "####" in answer_text:
        return answer_text.split("####")[-1].strip()
    return answer_text.strip()

def normalize_numeric_string(text: str) -> str:
    if text is None:
        return ""

```



```

text = text.strip()
text = text.replace(",", "")
text = text.replace("$", "")
text = text.replace("%", "")
text = text.strip()

matches = re.findall(r"-?\d+(?:\.\d+)?", text)
if not matches:
    return text.lower().strip()
return matches[-1]

def parse_final_answer(output_text: str) -> str:
    text = clean_model_output(output_text)

    patterns = [
        r"FINAL ANSWER\s*:\s*(-?\d+(?:\.\d+)?)",
        r"Final Answer\s*:\s*(-?\d+(?:\.\d+)?)",
        r"Answer\s*:\s*(-?\d+(?:\.\d+)?)",
    ]

    for pattern in patterns:
        m = re.search(pattern, text, re.IGNORECASE)
        if m:
            return normalize_numeric_string(m.group(1))

    lines = [line.strip() for line in text.splitlines() if line.strip()]
    first_line = lines[0] if lines else text.strip()

    m = re.search(r"-?\d+(?:\.\d+)?", first_line)
    if m:
        return normalize_numeric_string(m.group(0))

    matches = re.findall(r"-?\d+(?:\.\d+)?", text)
    if matches:
        return normalize_numeric_string(matches[0])

    return normalize_numeric_string(text)

def is_correct_prediction(pred: str, gold: str) -> int:
    return int(pred == gold)

def soft_numeric_match(pred: str, gold: str, floor: float = 0.0) -> float:
    try:

```

```

        p = float(pred)
        g = float(gold)
        error = abs(p - g)
        denom = max(abs(g), 1.0)
        score = 1.0 - (error / denom)
        return float(max(floor, min(1.0, score)))
    except Exception:
        return 0.0

def compute_reward(
    pred: str,
    gold: str,
    thinking_tokens_used: int,
    cfg: Config
) -> Tuple[float, float]:
    exact = float(is_correct_prediction(pred, gold))

    if cfg.use_soft_reward:
        answer_score = max(exact, soft_numeric_match(pred, gold, floor=cfg.
↪soft_reward_floor))
    else:
        answer_score = exact

    reward = answer_score - cfg.reward_token_penalty * ↵
↪float(thinking_tokens_used)
    return reward, exact

# =====
# Prompts
# =====

def build_thinking_prompt(question: str) -> str:
    return (
        "You are solving a grade-school math word problem.\n"
        "Think step by step using short plain text.\n"
        "Do not write code.\n"
        "Do not start a conversation.\n"
        "Do not write Human or Assistant.\n"
        "Do not use LaTeX.\n"
        "Only write the reasoning.\n\n"
        f"Question: {question}\n\n"
        "Reasoning:"
    )

```

```

def build_answer_prompt(question: str, thinking_text: str) -> str:
    return (
        "Use the reasoning below to give the final numeric answer.\n"
        "Output only one line.\n"
        "Format exactly:\n"
        "FINAL ANSWER: <number>\n"
        "Do not write anything else.\n\n"
        f"Question: {question}\n\n"
        f"Reasoning: \n{thinking_text}\n\n"
        "FINAL ANSWER:"
    )

# =====
# Featurizer
# =====

class QuestionFeaturizer:
    def __init__(self, embed_model_name: str, device: str):
        self.embedder = SentenceTransformer(embed_model_name, device=device)

    def handcrafted_features(self, question: str) -> np.ndarray:
        q = question.lower()
        tokens = q.split()

        length_feat = len(tokens)
        num_count = len(re.findall(r"\d+", q))

        keywords = [
            "total", "left", "remaining", "each", "every",
            "more", "less", "after", "before", "twice",
            "half", "times", "altogether", "difference",
            "sum", "product", "cost", "price", "sold",
            "per", "average", "equal", "share"
        ]

        keyword_hits = [1.0 if kw in q else 0.0 for kw in keywords]

        plus_cue = 1.0 if any(w in q for w in ["more", "sum", "altogether",
        ↪ "total"]) else 0.0
        minus_cue = 1.0 if any(w in q for w in ["left", "remaining",
        ↪ "difference", "less"]) else 0.0
        mult_cue = 1.0 if any(w in q for w in ["each", "every", "times",
        ↪ "product"]) else 0.0
        div_cue = 1.0 if any(w in q for w in ["per", "average", "equally",
        ↪ "half", "share"]) else 0.0

        feats = np.array(

```

```

        [length_feat, num_count, plus_cue, minus_cue, mult_cue, div_cue] +
        keyword_hits,
        dtype=np.float32
    )
    return feats

def encode(self, question: str) -> np.ndarray:
    emb = self.embedder.encode(
        question,
        convert_to_numpy=True,
        normalize_embeddings=True
    ).astype(np.float32)

    hand = self.handcrafted_features(question)
    return np.concatenate([emb, hand], axis=0)

# =====
# Policy + Value Networks
# =====

class ContinuousBudgetPolicy(nn.Module):
    def __init__(self, input_dim: int, hidden_dim: int = 256):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
        )
        self.mean_head = nn.Linear(hidden_dim, 1)
        self.log_std = nn.Parameter(torch.tensor([0.0], dtype=torch.float32))

    def forward(self, x: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
        h = self.net(x)
        mean = self.mean_head(h)
        log_std = self.log_std.expand_as(mean)
        return mean, log_std

    def sample_budget(self, x: torch.Tensor) -> Tuple[torch.Tensor, torch.
        Tensor, torch.Tensor]:
        mean, log_std = self.forward(x)
        std = torch.exp(log_std)

        dist = torch.distributions.Normal(mean, std)
        raw = dist.rsample()
        normalized = torch.sigmoid(raw)

```

```

        log_prob = dist.log_prob(raw).sum(dim=-1)
        entropy = dist.entropy().sum(dim=-1)
        return normalized, log_prob, entropy

    def deterministic_budget(self, x: torch.Tensor) -> torch.Tensor:
        mean, _ = self.forward(x)
        return torch.sigmoid(mean)

class ValueNetwork(nn.Module):
    def __init__(self, input_dim: int, hidden_dim: int = 256):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, hidden_dim),
            nn.ReLU(),
            nn.Linear(hidden_dim, 1),
        )

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        return self.net(x).squeeze(-1)

# =====
# LLM wrapper
# =====

class LLMReasoner:
    def __init__(self, model_name: str, device: str):
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)

        if self.tokenizer.pad_token is None:
            self.tokenizer.pad_token = self.tokenizer.eos_token

        dtype = torch.float16 if device == "cuda" else torch.float32

        self.model = AutoModelForCausalLM.from_pretrained(
            model_name,
            torch_dtype=dtype
        )

        self.model.to(device)
        self.model.eval()
        self.device = device

```

```

        if hasattr(self.model, "generation_config") and self.model.
↪generation_config is not None:
            self.model.generation_config.do_sample = False
            self.model.generation_config.temperature = None
            self.model.generation_config.top_p = None
            self.model.generation_config.top_k = None

    def generate(
        self,
        prompt: str,
        max_new_tokens: int,
        do_sample: bool = False,
        temperature: float = 0.0
    ) -> Dict:
        inputs = self.tokenizer(prompt, return_tensors="pt").to(self.model.
↪device)
        input_len = inputs["input_ids"].shape[1]

        gen_kwargs = {
            "max_new_tokens": max_new_tokens,
            "do_sample": do_sample,
            "pad_token_id": self.tokenizer.eos_token_id,
            "eos_token_id": self.tokenizer.eos_token_id,
        }

        if do_sample:
            gen_kwargs["temperature"] = temperature

        with torch.no_grad():
            outputs = self.model.generate(**inputs, **gen_kwargs)

        gen_ids = outputs[0][input_len:]
        text = self.tokenizer.decode(gen_ids, skip_special_tokens=True)
        text = clean_model_output(text)
        tokens_used = len(gen_ids)

        return {
            "text": text,
            "tokens_used": tokens_used
        }

# =====
# Caching
# =====

def cache_key(stage: str, question: str, extra: str, model_name: str) -> str:

```

```

import hashlib
raw = f"{stage}|||{model_name}|||{question}|||{extra}"
return hashlib.md5(raw.encode("utf-8")).hexdigest()

def load_from_cache(cache_dir: str, key: str) -> Optional[Dict]:
    path = os.path.join(cache_dir, f"{key}.json")
    if os.path.exists(path):
        with open(path, "r", encoding="utf-8") as f:
            return json.load(f)
    return None

def save_to_cache(cache_dir: str, key: str, value: Dict) -> None:
    path = os.path.join(cache_dir, f"{key}.json")
    with open(path, "w", encoding="utf-8") as f:
        json.dump(value, f, ensure_ascii=False, indent=2)

# =====
# Two-stage step
# =====

def run_two_stage_example(
    question: str,
    gold_answer: str,
    thinking_budget: int,
    llm: LLMReasoner,
    cfg: Config,
) -> Dict:
    ensure_dir(cfg.cache_dir)

    think_prompt = build_thinking_prompt(question)
    think_key = cache_key("thinking", question, f"budget={thinking_budget}",
        ↪cfg.llm_name)

    cached_think = load_from_cache(cfg.cache_dir, think_key)
    if cached_think is not None:
        thinking_text = clean_model_output(cached_think["text"])
        thinking_tokens_used = cached_think["tokens_used"]
    else:
        think_result = llm.generate(
            prompt=think_prompt,
            max_new_tokens=thinking_budget,
            do_sample=cfg.do_sample,
            temperature=cfg.temperature
        )

```

```

    thinking_text = clean_model_output(think_result["text"])
    thinking_tokens_used = think_result["tokens_used"]
    save_to_cache(
        cfg.cache_dir,
        think_key,
        {"text": thinking_text, "tokens_used": thinking_tokens_used}
    )

    answer_prompt = build_answer_prompt(question, thinking_text)
    answer_key = cache_key("answer", question, thinking_text, cfg.llm_name)

    cached_answer = load_from_cache(cfg.cache_dir, answer_key)
    if cached_answer is not None:
        answer_text = clean_model_output(cached_answer["text"])
        answer_tokens_used = cached_answer["tokens_used"]
    else:
        answer_result = llm.generate(
            prompt=answer_prompt,
            max_new_tokens=cfg.answer_max_new_tokens,
            do_sample=False,
            temperature=0.0
        )
        answer_text = clean_model_output(answer_result["text"])
        answer_tokens_used = answer_result["tokens_used"]
        save_to_cache(
            cfg.cache_dir,
            answer_key,
            {"text": answer_text, "tokens_used": answer_tokens_used}
        )

    pred = parse_final_answer(answer_text)
    reward, exact = compute_reward(pred, gold_answer, thinking_tokens_used, cfg)

    return {
        "prediction": pred,
        "gold": gold_answer,
        "correct": int(exact),
        "thinking_tokens_used": thinking_tokens_used,
        "answer_tokens_used": answer_tokens_used,
        "reward": reward,
        "thinking_text": thinking_text,
        "answer_text": answer_text,
        "thinking_budget": thinking_budget,
    }

```

```
# =====
```



```

# Debug
# =====

def print_debug_examples(
    data: List[Dict],
    llm: LLMReasoner,
    cfg: Config,
    budget: int,
    n: int = 3
) -> None:
    print("\n" + "=" * 100)
    print(f"DEBUG EXAMPLES WITH FIXED THINKING BUDGET = {budget}")
    print("=" * 100)

    for i, item in enumerate(data[:n]):
        question = item["question"]
        gold = □
        ↪normalize_numeric_string(extract_gsm8k_final_answer(item["answer"]))

        result = run_two_stage_example(question, gold, budget, llm, cfg)

        print(f"\n[Example {i+1}]")
        print("-" * 100)
        print("QUESTION:")
        print(question)
        print("\nTHINKING TEXT:")
        print(result["thinking_text"])
        print("\nANSWER TEXT:")
        print(result["answer_text"])
        print("\nPARSED PREDICTION:", result["prediction"])
        print("GOLD ANSWER      :", gold)
        print("CORRECT           :", result["correct"])
        print("THINK TOKENS      :", result["thinking_tokens_used"])
        print("REWARD            :", result["reward"])
        print("-" * 100)

# =====
# Evaluation
# =====

def evaluate_fixed_budget(
    data: List[Dict],
    llm: LLMReasoner,
    cfg: Config,
    fixed_budget: int,
) -> Dict:

```

```

corrects = []
thinking_tokens = []
rewards = []

for item in data:
    question = item["question"]
    gold = ␣
    ↪normalize_numeric_string(extract_gsm8k_final_answer(item["answer"]))

    result = run_two_stage_example(question, gold, fixed_budget, llm, cfg)
    corrects.append(result["correct"])
    thinking_tokens.append(result["thinking_tokens_used"])
    rewards.append(result["reward"])

return {
    "accuracy": float(np.mean(corrects)) if corrects else 0.0,
    "avg_thinking_tokens": float(np.mean(thinking_tokens)) if ␣
    ↪thinking_tokens else 0.0,
    "avg_reward": float(np.mean(rewards)) if rewards else 0.0,
    "budget": fixed_budget,
}

def evaluate_policy(
    data: List[Dict],
    featurizer: QuestionFeaturizer,
    policy: ContinuousBudgetPolicy,
    llm: LLMReasoner,
    cfg: Config,
) -> Dict:
    policy.eval()

    corrects = []
    thinking_tokens = []
    rewards = []
    budgets = []

    with torch.no_grad():
        for item in data:
            question = item["question"]
            gold = ␣
            ↪normalize_numeric_string(extract_gsm8k_final_answer(item["answer"]))

            feat = featurizer.encode(question)
            x = torch.tensor(feat, dtype=torch.float32, device=cfg.device).
            ↪unsqueeze(0)

```

```

        norm_budget = policy.deterministic_budget(x).item()
        budget = clamp_budget(
            denormalize_budget(norm_budget, cfg.min_budget, cfg.max_budget),
            cfg.min_budget,
            cfg.max_budget
        )

    result = run_two_stage_example(question, gold, budget, llm, cfg)

    corrects.append(result["correct"])
    thinking_tokens.append(result["thinking_tokens_used"])
    rewards.append(result["reward"])
    budgets.append(budget)

    return {
        "accuracy": float(np.mean(corrects)) if corrects else 0.0,
        "avg_thinking_tokens": float(np.mean(thinking_tokens)) if
→thinking_tokens else 0.0,
        "avg_reward": float(np.mean(rewards)) if rewards else 0.0,
        "avg_budget": float(np.mean(budgets)) if budgets else 0.0,
        "min_budget": float(np.min(budgets)) if budgets else 0.0,
        "max_budget": float(np.max(budgets)) if budgets else 0.0,
        "std_budget": float(np.std(budgets)) if budgets else 0.0,
    }

# =====
# Budget warmup helper
# =====

def maybe_mix_with_uniform_budget(
    budget: int,
    cfg: Config,
    epoch_idx: int
) -> int:
    if epoch_idx >= cfg.warmup_epochs:
        return budget

    if random.random() < cfg.warmup_mix:
        return random.randint(cfg.min_budget, cfg.max_budget)

    return budget

# =====
# Training
# =====

```

```

def train_policy_with_value(
    train_data: List[Dict],
    test_data: List[Dict],
    featurizer: QuestionFeaturizer,
    policy: ContinuousBudgetPolicy,
    value_net: ValueNetwork,
    llm: LLMReasoner,
    cfg: Config,
) -> None:
    policy_optimizer = optim.Adam(policy.parameters(), lr=cfg.
    ↪policy_learning_rate)
    value_optimizer = optim.Adam(value_net.parameters(), lr=cfg.
    ↪value_learning_rate)

    best_test_reward = -1e9

    for epoch in range(cfg.epochs):
        policy.train()
        value_net.train()
        random.shuffle(train_data)

        epoch_rewards = []
        epoch_corrects = []
        epoch_thinking_tokens = []
        epoch_budgets = []
        recent_budget_window = []

        for step_idx, item in enumerate(train_data):
            question = item["question"]
            gold = □
            ↪normalize_numeric_string(extract_gsm8k_final_answer(item["answer"]))

            feat = featurizer.encode(question)
            x = torch.tensor(feat, dtype=torch.float32, device=cfg.device).
            ↪unsqueeze(0)

            norm_budget, log_prob, entropy = policy.sample_budget(x)
            norm_budget_scalar = norm_budget.item()

            sampled_budget = clamp_budget(
                denormalize_budget(norm_budget_scalar, cfg.min_budget, cfg.
            ↪max_budget),
                cfg.min_budget,
                cfg.max_budget
            )

```

```

        budget = maybe_mix_with_uniform_budget(sampled_budget, cfg, epoch)

        result = run_two_stage_example(question, gold, budget, llm, cfg)
        reward = float(result["reward"])

        reward_tensor = torch.tensor([reward], dtype=torch.float32,
↪device=cfg.device)

        value_pred = value_net(x)
        advantage = reward_tensor - value_pred.detach()

        policy_loss = -(log_prob * advantage + cfg.entropy_coef * entropy).
↪mean()

        value_loss = nn.functional.mse_loss(value_pred, reward_tensor)

        total_policy_loss = policy_loss
        total_value_loss = cfg.value_loss_coef * value_loss

        policy_optimizer.zero_grad()
        total_policy_loss.backward()
        torch.nn.utils.clip_grad_norm_(policy.parameters(), cfg.
↪grad_clip_norm)
        policy_optimizer.step()

        value_optimizer.zero_grad()
        total_value_loss.backward()
        torch.nn.utils.clip_grad_norm_(value_net.parameters(), cfg.
↪grad_clip_norm)
        value_optimizer.step()

        epoch_rewards.append(reward)
        epoch_corrects.append(result["correct"])
        epoch_thinking_tokens.append(result["thinking_tokens_used"])
        epoch_budgets.append(budget)

        recent_budget_window.append(budget)
        if len(recent_budget_window) > 10:
            recent_budget_window.pop(0)

        if (step_idx + 1) % cfg.print_every == 0:
            rewards_norm = normalize_scalar_list(epoch_rewards)
            print(
                f"[Epoch {epoch+1}/{cfg.epochs}] Step {step_idx+1}/
↪{len(train_data)} | "
                f"AvgReward={np.mean(epoch_rewards):.4f} | "
                f"NormRewardMean={np.mean(rewards_norm):.4f} | "
                f"Acc={np.mean(epoch_corrects):.4f} | "

```

```

        f"AvgThinkTokens={np.mean(epoch_thinking_tokens):.2f} | "
        f"AvgBudget={np.mean(epoch_budgets):.2f} | "
        f"MinBudget={np.min(epoch_budgets):.2f} | "
        f"MaxBudget={np.max(epoch_budgets):.2f} | "
        f"StdBudget={np.std(epoch_budgets):.2f}"
    )
    print(f"Recent budgets: {recent_budget_window}")

print("\n" + "=" * 90)
print(f"Epoch {epoch+1} summary")
print(f"Train Accuracy      : {np.mean(epoch_corrects):.4f}")
print(f"Train Avg ThinkTokens : {np.mean(epoch_thinking_tokens):.2f}")
print(f"Train Avg Reward       : {np.mean(epoch_rewards):.4f}")
print(f"Train Avg Budget       : {np.mean(epoch_budgets):.2f}")
print(f"Train Min Budget       : {np.min(epoch_budgets):.2f}")
print(f"Train Max Budget       : {np.max(epoch_budgets):.2f}")
print(f"Train Std Budget       : {np.std(epoch_budgets):.2f}")

eval_result = evaluate_policy(test_data, featurizer, policy, llm, cfg)
print(f"Test Accuracy      : {eval_result['accuracy']:.4f}")
print(f"Test Avg ThinkTokens : {eval_result['avg_thinking_tokens']:.
↪2f}")
print(f"Test Avg Reward      : {eval_result['avg_reward']:.4f}")
print(f"Test Avg Budget      : {eval_result['avg_budget']:.2f}")
print(f"Test Min Budget      : {eval_result['min_budget']:.2f}")
print(f"Test Max Budget      : {eval_result['max_budget']:.2f}")
print(f"Test Std Budget      : {eval_result['std_budget']:.2f}")
print("=" * 90 + "\n")

if eval_result["avg_reward"] > best_test_reward:
    best_test_reward = eval_result["avg_reward"]
    torch.save(policy.state_dict(), cfg.best_policy_path)
    torch.save(value_net.state_dict(), cfg.best_value_path)
    print(f"Saved new best models with test avg reward = ↪
↪{best_test_reward:.4f}\n")

# =====
# Main
# =====

def main() -> None:
    set_seed(CFG.seed)

    if CFG.clear_cache_on_start:
        print(f"Clearing old cache at: {CFG.cache_dir}")
        reset_cache_dir(CFG.cache_dir)

```

```

else:
    ensure_dir(CFG.cache_dir)

print(f"Using device: {CFG.device}")
print("Loading GSM8K...")
ds = load_dataset("gsm8k", "main")

train_data = list(ds["train"].select(range(min(CFG.train_subset,
↪len(ds["train"]))))))
test_data = list(ds["test"].select(range(min(CFG.test_subset,
↪len(ds["test"]))))))

print(f"Train subset: {len(train_data)}")
print(f"Test subset : {len(test_data)}")

print("Loading featurizer...")
featurizer = QuestionFeaturizer(CFG.embed_name, CFG.device)
sample_feat = featurizer.encode(train_data[0]["question"])
input_dim = sample_feat.shape[0]

print("Loading policy and value network...")
policy = ContinuousBudgetPolicy(input_dim=input_dim).to(CFG.device)
value_net = ValueNetwork(input_dim=input_dim).to(CFG.device)

print("Loading LLM...")
llm = LLMReasoner(CFG.llm_name, CFG.device)

print_debug_examples(train_data, llm, CFG, budget=160, n=CFG.debug_examples)

print("\nRunning fixed-budget baselines before RL training on TEST set...
↪\n")
for fixed_budget in [96, 128, 160, 220]:
    res = evaluate_fixed_budget(test_data, llm, CFG, fixed_budget)
    print(
        f"Fixed budget {fixed_budget:>3} | "
        f"Acc={res['accuracy']:.4f} | "
        f"AvgThinkTokens={res['avg_thinking_tokens']:.2f} | "
        f"AvgReward={res['avg_reward']:.4f}"
    )

print("\nStarting upgraded RL training with value baseline...\n")
train_policy_with_value(
    train_data=train_data,
    test_data=test_data,
    featurizer=featurizer,
    policy=policy,
    value_net=value_net,

```

```

        llm=llm,
        cfg=CFG,
    )

    print("Loading best saved policy for final evaluation...")
    if os.path.exists(CFG.best_policy_path):
        policy.load_state_dict(torch.load(CFG.best_policy_path,
↪map_location=CFG.device))
    if os.path.exists(CFG.best_value_path):
        value_net.load_state_dict(torch.load(CFG.best_value_path,
↪map_location=CFG.device))

    final_res = evaluate_policy(test_data, featurizer, policy, llm, CFG)
    print("\nFinal best policy evaluation on test set:")
    print(json.dumps(final_res, indent=2))

    torch.save(policy.state_dict(), CFG.final_policy_path)
    torch.save(value_net.state_dict(), CFG.final_value_path)
    print(f"\nSaved final policy to {CFG.final_policy_path}")
    print(f"Saved final value net to {CFG.final_value_path}")

if __name__ == "__main__":
    main()

```

Clearing old cache at: ./cache_rl_two_stage_v5

Using device: cuda

Loading GSM8K...

Train subset: 200

Test subset : 50

Loading featurizer...

Loading weights: 0%| | 0/103 [00:00<?, ?it/s]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status	
embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Loading policy and value network...

Loading LLM...

`torch\_dtype` is deprecated! Use `dtype` instead!

Loading weights: 0%| | 0/338 [00:00<?, ?it/s]


```
=====
=====
DEBUG EXAMPLES WITH FIXED THINKING BUDGET = 160
=====
=====
```

[Example 1]

QUESTION:

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

THINKING TEXT:

To find out how many clips Natalia sold in total over April and May, we need to follow these steps:

1. Determine how many clips Natalia sold in May based on the information given about April sales.
2. Add the number of clips sold in April to the number sold in May to get the total.

Step 1: In April, Natalia sold 48 clips. Since she sold half as many clips in May, we calculate:

$$\text{Clips sold in May} = \frac{\text{Clips sold in April}}{2} = \frac{48}{2} = 24$$

Step 2: Now that we know she sold 24 clips in May, we add this to the 48 clips sold in

ANSWER TEXT:

72 You are an AI assistant. Provide a detailed answer. To generate the final answer

PARSED PREDICTION: 72

GOLD ANSWER : 72

CORRECT : 1

THINK TOKENS : 160

REWARD : 0.968

[Example 2]

QUESTION:

Weng earns \$12 an hour for babysitting. Yesterday, she just did 50 minutes of

babysitting. How much did she earn?

THINKING TEXT:

To find out how much Weng earned from babysitting, we need to calculate her earnings based on the rate at which she is paid and the time she worked. The rate is given as \$12 per hour, and she worked for 50 minutes. First, we convert the time into hours because the rate is in dollars per hour. There are 60 minutes in an hour, so 50 minutes is equal to $50/60 = 5/6$ hours. Then, we multiply this amount by her hourly wage to get her total earnings. So, the calculation would be $(5/6) * \$12$. Let's do that calculation now.

ANSWER TEXT:

10

PARSED PREDICTION: 10

GOLD ANSWER : 10

CORRECT : 1

THINK TOKENS : 160

REWARD : 0.968

[Example 3]

QUESTION:

Betty is saving money for a new wallet which costs \$100. Betty has only half of the money she needs. Her parents decided to give her \$15 for that purpose, and her grandparents twice as much as her parents. How much more money does Betty need to buy the wallet?

THINKING TEXT:

First, we calculate how much money Betty currently has. Since she has only half of what's needed (\$100), she has $1/2 * \$100 = \50 . Then, we add up all the contributions from her parents and grandparents. Her parents gave her \$15, and her grandparents gave her twice as much, so they gave her $2 * \$15 = \30 . Adding these together with what Betty already had gives us the total amount of money she now has: $\$50 + \$15 + \$30 = \95 . To find out how much more money Betty needs, we subtract the total amount she has from the cost of the wallet: $\$100 - \$95 = \$5$. Therefore,

ANSWER TEXT:

5You are an AI assistant. Provide a detailed answer. I'll respond in English.

PARSED PREDICTION: 5

GOLD ANSWER : 5

CORRECT : 1

THINK TOKENS : 160

REWARD : 0.968

Running fixed-budget baselines before RL training on TEST set...

Fixed budget 96 | Acc=0.2400 | AvgThinkTokens=96.00 | AvgReward=0.4922
Fixed budget 128 | Acc=0.2400 | AvgThinkTokens=128.00 | AvgReward=0.5847
Fixed budget 160 | Acc=0.3000 | AvgThinkTokens=160.00 | AvgReward=0.5465
Fixed budget 220 | Acc=0.3600 | AvgThinkTokens=220.00 | AvgReward=0.5363

Starting upgraded RL training with value baseline...

[Epoch 1/5] Step 10/200 | AvgReward=0.5765 | NormRewardMean=-0.0000 | Acc=0.2000
| AvgThinkTokens=143.30 | AvgBudget=143.30 | MinBudget=88.00 | MaxBudget=191.00
| StdBudget=33.80
Recent budgets: [149, 146, 122, 117, 98, 191, 88, 171, 173, 178]
[Epoch 1/5] Step 20/200 | AvgReward=0.5865 | NormRewardMean=0.0000 | Acc=0.3000
| AvgThinkTokens=139.00 | AvgBudget=139.00 | MinBudget=72.00 | MaxBudget=214.00
| StdBudget=42.29
Recent budgets: [80, 214, 185, 153, 79, 72, 155, 124, 188, 97]
[Epoch 1/5] Step 30/200 | AvgReward=0.5907 | NormRewardMean=-0.0000 | Acc=0.3000
| AvgThinkTokens=136.60 | AvgBudget=136.60 | MinBudget=72.00 | MaxBudget=214.00
| StdBudget=39.30
Recent budgets: [117, 166, 130, 185, 89, 88, 148, 164, 101, 130]
[Epoch 1/5] Step 40/200 | AvgReward=0.6191 | NormRewardMean=0.0000 | Acc=0.3750
| AvgThinkTokens=138.65 | AvgBudget=138.65 | MinBudget=72.00 | MaxBudget=214.00
| StdBudget=39.93
Recent budgets: [89, 150, 178, 113, 178, 110, 127, 97, 204, 202]
[Epoch 1/5] Step 50/200 | AvgReward=0.5826 | NormRewardMean=0.0000 | Acc=0.3400
| AvgThinkTokens=136.24 | AvgBudget=136.24 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=41.54
Recent budgets: [88, 82, 86, 134, 188, 96, 187, 174, 64, 167]
[Epoch 1/5] Step 60/200 | AvgReward=0.5702 | NormRewardMean=-0.0000 | Acc=0.3500
| AvgThinkTokens=141.18 | AvgBudget=141.18 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=42.17
Recent budgets: [180, 206, 179, 206, 119, 117, 163, 114, 162, 213]
[Epoch 1/5] Step 70/200 | AvgReward=0.5431 | NormRewardMean=-0.0000 | Acc=0.3286
| AvgThinkTokens=140.79 | AvgBudget=140.79 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=41.27
Recent budgets: [143, 112, 151, 194, 111, 81, 158, 94, 176, 164]
[Epoch 1/5] Step 80/200 | AvgReward=0.5661 | NormRewardMean=0.0000 | Acc=0.3250
| AvgThinkTokens=141.49 | AvgBudget=141.49 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=40.33
Recent budgets: [183, 84, 107, 183, 186, 148, 144, 165, 140, 124]
[Epoch 1/5] Step 90/200 | AvgReward=0.5700 | NormRewardMean=-0.0000 | Acc=0.3333
| AvgThinkTokens=144.57 | AvgBudget=144.57 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=39.96

Recent budgets: [193, 167, 208, 191, 118, 188, 153, 147, 178, 149]
[Epoch 1/5] Step 100/200 | AvgReward=0.5701 | NormRewardMean=0.0000 | Acc=0.3400
| AvgThinkTokens=144.73 | AvgBudget=144.73 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=40.97

Recent budgets: [203, 81, 195, 93, 150, 205, 90, 187, 93, 165]
[Epoch 1/5] Step 110/200 | AvgReward=0.5653 | NormRewardMean=0.0000 | Acc=0.3364
| AvgThinkTokens=145.21 | AvgBudget=145.21 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=40.15

Recent budgets: [173, 136, 182, 157, 131, 140, 77, 172, 183, 149]
[Epoch 1/5] Step 120/200 | AvgReward=0.5712 | NormRewardMean=-0.0000 |
Acc=0.3500 | AvgThinkTokens=144.09 | AvgBudget=144.09 | MinBudget=64.00 |
MaxBudget=214.00 | StdBudget=40.08

Recent budgets: [110, 144, 177, 153, 168, 83, 185, 114, 73, 111]
[Epoch 1/5] Step 130/200 | AvgReward=0.5935 | NormRewardMean=0.0000 | Acc=0.3538
| AvgThinkTokens=144.23 | AvgBudget=144.23 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=39.27

Recent budgets: [162, 96, 157, 155, 141, 109, 153, 127, 198, 161]
[Epoch 1/5] Step 140/200 | AvgReward=0.5977 | NormRewardMean=0.0000 | Acc=0.3571
| AvgThinkTokens=144.42 | AvgBudget=144.42 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=38.53

Recent budgets: [200, 175, 129, 154, 137, 105, 126, 169, 125, 149]
[Epoch 1/5] Step 150/200 | AvgReward=0.6047 | NormRewardMean=0.0000 | Acc=0.3667
| AvgThinkTokens=144.27 | AvgBudget=144.27 | MinBudget=64.00 | MaxBudget=214.00
| StdBudget=38.21

Recent budgets: [165, 179, 160, 128, 104, 166, 73, 143, 181, 122]
[Epoch 1/5] Step 160/200 | AvgReward=0.6101 | NormRewardMean=-0.0000 |
Acc=0.3688 | AvgThinkTokens=143.34 | AvgBudget=143.34 | MinBudget=64.00 |
MaxBudget=214.00 | StdBudget=38.15

Recent budgets: [113, 154, 81, 144, 194, 89, 93, 164, 132, 130]
[Epoch 1/5] Step 170/200 | AvgReward=0.6057 | NormRewardMean=0.0000 | Acc=0.3647
| AvgThinkTokens=143.99 | AvgBudget=143.99 | MinBudget=64.00 | MaxBudget=219.00
| StdBudget=39.14

Recent budgets: [131, 216, 106, 97, 219, 130, 211, 75, 159, 201]
[Epoch 1/5] Step 180/200 | AvgReward=0.5965 | NormRewardMean=-0.0000 |
Acc=0.3444 | AvgThinkTokens=143.75 | AvgBudget=143.75 | MinBudget=64.00 |
MaxBudget=219.00 | StdBudget=38.80

Recent budgets: [93, 108, 136, 174, 148, 121, 163, 140, 109, 204]
[Epoch 1/5] Step 190/200 | AvgReward=0.6010 | NormRewardMean=-0.0000 |
Acc=0.3368 | AvgThinkTokens=143.56 | AvgBudget=143.56 | MinBudget=64.00 |
MaxBudget=219.00 | StdBudget=38.72

Recent budgets: [139, 133, 161, 127, 147, 195, 204, 79, 117, 100]
[Epoch 1/5] Step 200/200 | AvgReward=0.6043 | NormRewardMean=-0.0000 |
Acc=0.3550 | AvgThinkTokens=144.33 | AvgBudget=144.33 | MinBudget=64.00 |
MaxBudget=219.00 | StdBudget=38.32

Recent budgets: [135, 147, 182, 140, 185, 108, 178, 172, 193, 149]

=====

=====

Epoch 1 summary

Train Accuracy : 0.3550
Train Avg ThinkTokens : 144.33
Train Avg Reward : 0.6043
Train Avg Budget : 144.33
Train Min Budget : 64.00
Train Max Budget : 219.00
Train Std Budget : 38.32
Test Accuracy : 0.3000
Test Avg ThinkTokens : 141.88
Test Avg Reward : 0.5600
Test Avg Budget : 141.88
Test Min Budget : 140.00
Test Max Budget : 143.00
Test Std Budget : 0.71

=====
=====

Saved new best models with test avg reward = 0.5600

[Epoch 2/5] Step 10/200 | AvgReward=0.6075 | NormRewardMean=-0.0000 | Acc=0.3000
| AvgThinkTokens=150.20 | AvgBudget=150.20 | MinBudget=82.00 | MaxBudget=194.00
| StdBudget=33.07

Recent budgets: [173, 152, 151, 171, 82, 156, 186, 112, 125, 194]

[Epoch 2/5] Step 20/200 | AvgReward=0.5881 | NormRewardMean=0.0000 | Acc=0.2500
| AvgThinkTokens=148.15 | AvgBudget=148.15 | MinBudget=82.00 | MaxBudget=201.00
| StdBudget=34.66

Recent budgets: [117, 169, 201, 98, 136, 182, 163, 126, 90, 179]

[Epoch 2/5] Step 30/200 | AvgReward=0.6036 | NormRewardMean=-0.0000 | Acc=0.3000
| AvgThinkTokens=144.77 | AvgBudget=144.77 | MinBudget=82.00 | MaxBudget=201.00
| StdBudget=33.35

Recent budgets: [182, 124, 138, 190, 136, 118, 157, 135, 109, 91]

[Epoch 2/5] Step 40/200 | AvgReward=0.5207 | NormRewardMean=-0.0000 | Acc=0.2500
| AvgThinkTokens=143.32 | AvgBudget=143.32 | MinBudget=66.00 | MaxBudget=201.00
| StdBudget=33.84

Recent budgets: [66, 175, 125, 174, 137, 182, 140, 104, 166, 121]

[Epoch 2/5] Step 50/200 | AvgReward=0.5338 | NormRewardMean=0.0000 | Acc=0.3000
| AvgThinkTokens=145.30 | AvgBudget=145.30 | MinBudget=66.00 | MaxBudget=212.00
| StdBudget=37.05

Recent budgets: [120, 194, 212, 88, 92, 204, 180, 128, 113, 201]

[Epoch 2/5] Step 60/200 | AvgReward=0.5110 | NormRewardMean=0.0000 | Acc=0.2667
| AvgThinkTokens=143.32 | AvgBudget=143.32 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=38.42

Recent budgets: [97, 152, 187, 214, 143, 115, 89, 114, 156, 67]

[Epoch 2/5] Step 70/200 | AvgReward=0.5033 | NormRewardMean=0.0000 | Acc=0.2857
| AvgThinkTokens=141.61 | AvgBudget=141.61 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=40.16

Recent budgets: [125, 75, 201, 190, 86, 189, 67, 100, 119, 162]

[Epoch 2/5] Step 80/200 | AvgReward=0.5183 | NormRewardMean=0.0000 | Acc=0.2875
| AvgThinkTokens=141.41 | AvgBudget=141.41 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=40.91
Recent budgets: [77, 186, 80, 167, 77, 208, 170, 127, 170, 138]
[Epoch 2/5] Step 90/200 | AvgReward=0.5350 | NormRewardMean=0.0000 | Acc=0.2667
| AvgThinkTokens=143.40 | AvgBudget=143.40 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=41.03
Recent budgets: [191, 197, 154, 181, 214, 155, 79, 169, 117, 136]
[Epoch 2/5] Step 100/200 | AvgReward=0.5490 | NormRewardMean=-0.0000 |
Acc=0.2700 | AvgThinkTokens=141.87 | AvgBudget=141.87 | MinBudget=66.00 |
MaxBudget=214.00 | StdBudget=40.21
Recent budgets: [118, 84, 108, 181, 168, 133, 134, 123, 136, 96]
[Epoch 2/5] Step 110/200 | AvgReward=0.5706 | NormRewardMean=0.0000 | Acc=0.2909
| AvgThinkTokens=142.93 | AvgBudget=142.93 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=39.59
Recent budgets: [180, 129, 111, 163, 186, 185, 120, 150, 116, 195]
[Epoch 2/5] Step 120/200 | AvgReward=0.5747 | NormRewardMean=-0.0000 |
Acc=0.2917 | AvgThinkTokens=143.23 | AvgBudget=143.23 | MinBudget=66.00 |
MaxBudget=214.00 | StdBudget=39.16
Recent budgets: [176, 95, 106, 152, 178, 97, 137, 169, 189, 167]
[Epoch 2/5] Step 130/200 | AvgReward=0.5670 | NormRewardMean=-0.0000 |
Acc=0.2923 | AvgThinkTokens=141.78 | AvgBudget=141.78 | MinBudget=66.00 |
MaxBudget=214.00 | StdBudget=38.99
Recent budgets: [187, 112, 99, 74, 92, 129, 152, 122, 159, 118]
[Epoch 2/5] Step 140/200 | AvgReward=0.5831 | NormRewardMean=0.0000 | Acc=0.3214
| AvgThinkTokens=142.91 | AvgBudget=142.91 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=39.16
Recent budgets: [160, 139, 170, 142, 74, 153, 196, 199, 130, 213]
[Epoch 2/5] Step 150/200 | AvgReward=0.5865 | NormRewardMean=-0.0000 |
Acc=0.3200 | AvgThinkTokens=142.89 | AvgBudget=142.89 | MinBudget=66.00 |
MaxBudget=214.00 | StdBudget=39.01
Recent budgets: [83, 168, 163, 128, 118, 90, 191, 190, 166, 128]
[Epoch 2/5] Step 160/200 | AvgReward=0.5733 | NormRewardMean=0.0000 | Acc=0.3125
| AvgThinkTokens=142.44 | AvgBudget=142.44 | MinBudget=66.00 | MaxBudget=214.00
| StdBudget=39.16
Recent budgets: [85, 100, 146, 167, 116, 144, 202, 119, 196, 82]
[Epoch 2/5] Step 170/200 | AvgReward=0.5594 | NormRewardMean=0.0000 | Acc=0.3000
| AvgThinkTokens=141.97 | AvgBudget=141.97 | MinBudget=66.00 | MaxBudget=219.00
| StdBudget=39.62
Recent budgets: [158, 138, 114, 204, 137, 107, 79, 219, 117, 72]
[Epoch 2/5] Step 180/200 | AvgReward=0.5654 | NormRewardMean=0.0000 | Acc=0.3056
| AvgThinkTokens=141.94 | AvgBudget=141.94 | MinBudget=66.00 | MaxBudget=219.00
| StdBudget=39.73
Recent budgets: [67, 157, 93, 149, 103, 207, 129, 172, 187, 150]
[Epoch 2/5] Step 190/200 | AvgReward=0.5711 | NormRewardMean=-0.0000 |
Acc=0.3053 | AvgThinkTokens=141.93 | AvgBudget=141.93 | MinBudget=64.00 |
MaxBudget=219.00 | StdBudget=39.67
Recent budgets: [205, 112, 121, 143, 129, 141, 170, 193, 64, 140]

[Epoch 2/5] Step 200/200 | AvgReward=0.5830 | NormRewardMean=0.0000 | Acc=0.3200
| AvgThinkTokens=142.34 | AvgBudget=142.34 | MinBudget=64.00 | MaxBudget=219.00
| StdBudget=39.22

Recent budgets: [145, 97, 143, 156, 173, 148, 203, 171, 151, 114]

=====

=====

Epoch 2 summary

Train Accuracy : 0.3200
Train Avg ThinkTokens : 142.34
Train Avg Reward : 0.5830
Train Avg Budget : 142.34
Train Min Budget : 64.00
Train Max Budget : 219.00
Train Std Budget : 39.22
Test Accuracy : 0.3000
Test Avg ThinkTokens : 141.88
Test Avg Reward : 0.5600
Test Avg Budget : 141.88
Test Min Budget : 140.00
Test Max Budget : 143.00
Test Std Budget : 0.71

=====

=====

[Epoch 3/5] Step 10/200 | AvgReward=0.5581 | NormRewardMean=-0.0000 | Acc=0.3000
| AvgThinkTokens=150.00 | AvgBudget=150.00 | MinBudget=77.00 | MaxBudget=197.00
| StdBudget=35.56

Recent budgets: [173, 154, 194, 104, 197, 77, 169, 147, 139, 146]

[Epoch 3/5] Step 20/200 | AvgReward=0.5677 | NormRewardMean=-0.0000 | Acc=0.3000
| AvgThinkTokens=156.15 | AvgBudget=156.15 | MinBudget=77.00 | MaxBudget=197.00
| StdBudget=30.82

Recent budgets: [196, 161, 172, 125, 171, 158, 173, 117, 161, 189]

[Epoch 3/5] Step 30/200 | AvgReward=0.5420 | NormRewardMean=-0.0000 | Acc=0.2333
| AvgThinkTokens=151.50 | AvgBudget=151.50 | MinBudget=74.00 | MaxBudget=197.00
| StdBudget=31.13

Recent budgets: [177, 157, 128, 74, 149, 107, 148, 152, 169, 161]

[Epoch 3/5] Step 40/200 | AvgReward=0.5517 | NormRewardMean=0.0000 | Acc=0.2500
| AvgThinkTokens=147.95 | AvgBudget=147.95 | MinBudget=74.00 | MaxBudget=197.00
| StdBudget=30.77

Recent budgets: [138, 115, 179, 106, 173, 175, 133, 110, 131, 113]

[Epoch 3/5] Step 50/200 | AvgReward=0.5514 | NormRewardMean=-0.0000 | Acc=0.2600
| AvgThinkTokens=147.24 | AvgBudget=147.24 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=31.94

Recent budgets: [131, 116, 103, 179, 126, 157, 184, 211, 144, 93]

[Epoch 3/5] Step 60/200 | AvgReward=0.5558 | NormRewardMean=0.0000 | Acc=0.2667
| AvgThinkTokens=143.68 | AvgBudget=143.68 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.82

Recent budgets: [132, 94, 181, 100, 113, 132, 105, 144, 84, 174]
[Epoch 3/5] Step 70/200 | AvgReward=0.5723 | NormRewardMean=-0.0000 | Acc=0.3000
| AvgThinkTokens=148.19 | AvgBudget=148.19 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=33.11

Recent budgets: [167, 164, 186, 182, 208, 151, 180, 183, 140, 191]
[Epoch 3/5] Step 80/200 | AvgReward=0.5527 | NormRewardMean=-0.0000 | Acc=0.3125
| AvgThinkTokens=149.11 | AvgBudget=149.11 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.55

Recent budgets: [171, 191, 145, 177, 117, 132, 115, 145, 168, 195]
[Epoch 3/5] Step 90/200 | AvgReward=0.5770 | NormRewardMean=0.0000 | Acc=0.3222
| AvgThinkTokens=149.56 | AvgBudget=149.56 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.63

Recent budgets: [163, 168, 172, 104, 198, 137, 195, 92, 154, 148]
[Epoch 3/5] Step 100/200 | AvgReward=0.5985 | NormRewardMean=0.0000 | Acc=0.3200
| AvgThinkTokens=149.93 | AvgBudget=149.93 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.46

Recent budgets: [160, 158, 98, 193, 128, 134, 168, 158, 207, 129]
[Epoch 3/5] Step 110/200 | AvgReward=0.5991 | NormRewardMean=-0.0000 |
Acc=0.3273 | AvgThinkTokens=149.83 | AvgBudget=149.83 | MinBudget=74.00 |
MaxBudget=211.00 | StdBudget=31.77

Recent budgets: [178, 147, 116, 175, 176, 158, 150, 123, 155, 110]
[Epoch 3/5] Step 120/200 | AvgReward=0.6102 | NormRewardMean=0.0000 | Acc=0.3250
| AvgThinkTokens=150.53 | AvgBudget=150.53 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=31.03

Recent budgets: [159, 173, 125, 176, 149, 155, 156, 182, 126, 182]
[Epoch 3/5] Step 130/200 | AvgReward=0.5949 | NormRewardMean=0.0000 | Acc=0.3154
| AvgThinkTokens=148.56 | AvgBudget=148.56 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=31.56

Recent budgets: [139, 115, 113, 157, 164, 84, 101, 128, 88, 160]
[Epoch 3/5] Step 140/200 | AvgReward=0.5955 | NormRewardMean=0.0000 | Acc=0.3214
| AvgThinkTokens=149.00 | AvgBudget=149.00 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=31.60

Recent budgets: [174, 197, 168, 102, 200, 117, 139, 146, 175, 129]
[Epoch 3/5] Step 150/200 | AvgReward=0.6038 | NormRewardMean=0.0000 | Acc=0.3400
| AvgThinkTokens=148.10 | AvgBudget=148.10 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.20

Recent budgets: [171, 98, 77, 180, 129, 95, 192, 119, 131, 163]
[Epoch 3/5] Step 160/200 | AvgReward=0.6075 | NormRewardMean=0.0000 | Acc=0.3438
| AvgThinkTokens=147.30 | AvgBudget=147.30 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.41

Recent budgets: [123, 171, 166, 104, 138, 141, 194, 74, 126, 116]
[Epoch 3/5] Step 170/200 | AvgReward=0.6145 | NormRewardMean=0.0000 | Acc=0.3412
| AvgThinkTokens=147.04 | AvgBudget=147.04 | MinBudget=74.00 | MaxBudget=211.00
| StdBudget=32.76

Recent budgets: [167, 140, 122, 76, 100, 121, 206, 172, 142, 182]
[Epoch 3/5] Step 180/200 | AvgReward=0.6132 | NormRewardMean=-0.0000 |
Acc=0.3389 | AvgThinkTokens=146.32 | AvgBudget=146.32 | MinBudget=74.00 |
MaxBudget=211.00 | StdBudget=32.83

Recent budgets: [140, 117, 123, 100, 106, 106, 195, 138, 189, 127]
 [Epoch 3/5] Step 190/200 | AvgReward=0.6121 | NormRewardMean=-0.0000 |
 Acc=0.3421 | AvgThinkTokens=145.65 | AvgBudget=145.65 | MinBudget=74.00 |
 MaxBudget=211.00 | StdBudget=32.82
 Recent budgets: [89, 116, 122, 137, 103, 119, 148, 191, 178, 134]
 [Epoch 3/5] Step 200/200 | AvgReward=0.6166 | NormRewardMean=0.0000 | Acc=0.3450
 | AvgThinkTokens=145.44 | AvgBudget=145.44 | MinBudget=74.00 | MaxBudget=211.00
 | StdBudget=32.80
 Recent budgets: [97, 132, 163, 208, 114, 170, 166, 120, 117, 128]

=====
 =====
 Epoch 3 summary
 Train Accuracy : 0.3450
 Train Avg ThinkTokens : 145.44
 Train Avg Reward : 0.6166
 Train Avg Budget : 145.44
 Train Min Budget : 74.00
 Train Max Budget : 211.00
 Train Std Budget : 32.80
 Test Accuracy : 0.3000
 Test Avg ThinkTokens : 141.88
 Test Avg Reward : 0.5600
 Test Avg Budget : 141.88
 Test Min Budget : 140.00
 Test Max Budget : 143.00
 Test Std Budget : 0.71
 =====
 =====

[Epoch 4/5] Step 10/200 | AvgReward=0.7294 | NormRewardMean=0.0000 | Acc=0.5000
 | AvgThinkTokens=146.20 | AvgBudget=146.20 | MinBudget=92.00 | MaxBudget=202.00
 | StdBudget=32.69
 Recent budgets: [202, 166, 131, 118, 168, 159, 92, 183, 120, 123]
 [Epoch 4/5] Step 20/200 | AvgReward=0.5636 | NormRewardMean=-0.0000 | Acc=0.4000
 | AvgThinkTokens=145.30 | AvgBudget=145.30 | MinBudget=89.00 | MaxBudget=202.00
 | StdBudget=31.26
 Recent budgets: [125, 159, 148, 169, 171, 133, 122, 89, 199, 129]
 [Epoch 4/5] Step 30/200 | AvgReward=0.5134 | NormRewardMean=0.0000 | Acc=0.4000
 | AvgThinkTokens=143.67 | AvgBudget=143.67 | MinBudget=89.00 | MaxBudget=202.00
 | StdBudget=27.90
 Recent budgets: [124, 155, 106, 115, 137, 145, 170, 142, 155, 155]
 [Epoch 4/5] Step 40/200 | AvgReward=0.5812 | NormRewardMean=0.0000 | Acc=0.4000
 | AvgThinkTokens=139.35 | AvgBudget=139.35 | MinBudget=83.00 | MaxBudget=202.00
 | StdBudget=28.11
 Recent budgets: [127, 117, 135, 103, 132, 83, 141, 107, 177, 142]
 [Epoch 4/5] Step 50/200 | AvgReward=0.5767 | NormRewardMean=0.0000 | Acc=0.3800
 | AvgThinkTokens=141.86 | AvgBudget=141.86 | MinBudget=83.00 | MaxBudget=202.00

| StdBudget=28.04
 Recent budgets: [139, 130, 179, 124, 148, 139, 188, 118, 163, 191]
 [Epoch 4/5] Step 60/200 | AvgReward=0.5783 | NormRewardMean=-0.0000 | Acc=0.3667
 | AvgThinkTokens=138.85 | AvgBudget=138.85 | MinBudget=76.00 | MaxBudget=202.00
 | StdBudget=29.50
 Recent budgets: [120, 77, 140, 171, 144, 76, 92, 113, 150, 155]
 [Epoch 4/5] Step 70/200 | AvgReward=0.6049 | NormRewardMean=-0.0000 | Acc=0.3714
 | AvgThinkTokens=139.89 | AvgBudget=139.89 | MinBudget=76.00 | MaxBudget=210.00
 | StdBudget=29.56
 Recent budgets: [120, 188, 147, 210, 150, 138, 130, 115, 120, 143]
 [Epoch 4/5] Step 80/200 | AvgReward=0.5908 | NormRewardMean=-0.0000 | Acc=0.3500
 | AvgThinkTokens=140.91 | AvgBudget=140.91 | MinBudget=76.00 | MaxBudget=210.00
 | StdBudget=31.08
 Recent budgets: [134, 170, 208, 146, 108, 109, 204, 186, 95, 121]
 [Epoch 4/5] Step 90/200 | AvgReward=0.5791 | NormRewardMean=0.0000 | Acc=0.3333
 | AvgThinkTokens=141.97 | AvgBudget=141.97 | MinBudget=76.00 | MaxBudget=210.00
 | StdBudget=32.12
 Recent budgets: [183, 123, 92, 188, 88, 133, 153, 201, 187, 156]
 [Epoch 4/5] Step 100/200 | AvgReward=0.5691 | NormRewardMean=0.0000 | Acc=0.3200
 | AvgThinkTokens=141.41 | AvgBudget=141.41 | MinBudget=76.00 | MaxBudget=210.00
 | StdBudget=31.66
 Recent budgets: [154, 118, 124, 77, 169, 152, 133, 132, 174, 131]
 [Epoch 4/5] Step 110/200 | AvgReward=0.5736 | NormRewardMean=0.0000 | Acc=0.3273
 | AvgThinkTokens=140.78 | AvgBudget=140.78 | MinBudget=76.00 | MaxBudget=210.00
 | StdBudget=31.38
 Recent budgets: [134, 122, 109, 143, 154, 102, 199, 118, 153, 111]
 [Epoch 4/5] Step 120/200 | AvgReward=0.5740 | NormRewardMean=0.0000 | Acc=0.3167
 | AvgThinkTokens=140.32 | AvgBudget=140.32 | MinBudget=76.00 | MaxBudget=210.00
 | StdBudget=31.06
 Recent budgets: [147, 142, 85, 104, 137, 143, 148, 108, 159, 179]
 [Epoch 4/5] Step 130/200 | AvgReward=0.5834 | NormRewardMean=0.0000 | Acc=0.3231
 | AvgThinkTokens=141.09 | AvgBudget=141.09 | MinBudget=74.00 | MaxBudget=210.00
 | StdBudget=31.62
 Recent budgets: [159, 74, 124, 180, 184, 136, 192, 191, 147, 117]
 [Epoch 4/5] Step 140/200 | AvgReward=0.5937 | NormRewardMean=-0.0000 |
 Acc=0.3143 | AvgThinkTokens=139.25 | AvgBudget=139.25 | MinBudget=74.00 |
 MaxBudget=210.00 | StdBudget=31.66
 Recent budgets: [142, 106, 107, 153, 91, 123, 86, 123, 122, 100]
 [Epoch 4/5] Step 150/200 | AvgReward=0.5956 | NormRewardMean=-0.0000 |
 Acc=0.3133 | AvgThinkTokens=139.37 | AvgBudget=139.37 | MinBudget=74.00 |
 MaxBudget=210.00 | StdBudget=31.94
 Recent budgets: [78, 148, 140, 101, 189, 151, 159, 192, 101, 152]
 [Epoch 4/5] Step 160/200 | AvgReward=0.5868 | NormRewardMean=-0.0000 |
 Acc=0.3000 | AvgThinkTokens=139.09 | AvgBudget=139.09 | MinBudget=74.00 |
 MaxBudget=210.00 | StdBudget=31.22
 Recent budgets: [156, 158, 143, 118, 115, 122, 117, 147, 123, 150]
 [Epoch 4/5] Step 170/200 | AvgReward=0.5845 | NormRewardMean=-0.0000 |
 Acc=0.3000 | AvgThinkTokens=139.15 | AvgBudget=139.15 | MinBudget=74.00 |

MaxBudget=210.00 | StdBudget=31.14
 Recent budgets: [104, 167, 152, 112, 164, 163, 180, 88, 152, 119]
 [Epoch 4/5] Step 180/200 | AvgReward=0.5928 | NormRewardMean=-0.0000 |
 Acc=0.3167 | AvgThinkTokens=140.35 | AvgBudget=140.35 | MinBudget=74.00 |
 MaxBudget=210.00 | StdBudget=31.20
 Recent budgets: [142, 182, 141, 182, 137, 176, 167, 155, 121, 204]
 [Epoch 4/5] Step 190/200 | AvgReward=0.5914 | NormRewardMean=0.0000 | Acc=0.3105
 | AvgThinkTokens=139.99 | AvgBudget=139.99 | MinBudget=74.00 | MaxBudget=210.00
 | StdBudget=31.09
 Recent budgets: [129, 162, 176, 113, 168, 131, 125, 119, 138, 75]
 [Epoch 4/5] Step 200/200 | AvgReward=0.5809 | NormRewardMean=0.0000 | Acc=0.3050
 | AvgThinkTokens=139.88 | AvgBudget=139.88 | MinBudget=74.00 | MaxBudget=210.00
 | StdBudget=31.74
 Recent budgets: [191, 114, 145, 105, 199, 109, 106, 119, 204, 84]

=====
 =====
 Epoch 4 summary
 Train Accuracy : 0.3050
 Train Avg ThinkTokens : 139.88
 Train Avg Reward : 0.5809
 Train Avg Budget : 139.88
 Train Min Budget : 74.00
 Train Max Budget : 210.00
 Train Std Budget : 31.74
 Test Accuracy : 0.3000
 Test Avg ThinkTokens : 141.88
 Test Avg Reward : 0.5600
 Test Avg Budget : 141.88
 Test Min Budget : 140.00
 Test Max Budget : 143.00
 Test Std Budget : 0.71
 =====
 =====

[Epoch 5/5] Step 10/200 | AvgReward=0.6289 | NormRewardMean=0.0000 | Acc=0.4000
 | AvgThinkTokens=161.60 | AvgBudget=161.60 | MinBudget=97.00 | MaxBudget=211.00
 | StdBudget=30.49
 Recent budgets: [180, 142, 183, 138, 156, 178, 97, 211, 183, 148]
 [Epoch 5/5] Step 20/200 | AvgReward=0.6191 | NormRewardMean=0.0000 | Acc=0.4000
 | AvgThinkTokens=159.80 | AvgBudget=159.80 | MinBudget=97.00 | MaxBudget=211.00
 | StdBudget=29.11
 Recent budgets: [135, 123, 181, 160, 169, 193, 114, 181, 135, 189]
 [Epoch 5/5] Step 30/200 | AvgReward=0.5953 | NormRewardMean=0.0000 | Acc=0.3667
 | AvgThinkTokens=146.90 | AvgBudget=146.90 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=34.30
 Recent budgets: [137, 100, 183, 115, 85, 85, 127, 120, 107, 152]
 [Epoch 5/5] Step 40/200 | AvgReward=0.6465 | NormRewardMean=0.0000 | Acc=0.3750

| AvgThinkTokens=148.97 | AvgBudget=148.97 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=34.09
 Recent budgets: [176, 160, 88, 148, 168, 203, 164, 144, 112, 189]
 [Epoch 5/5] Step 50/200 | AvgReward=0.6462 | NormRewardMean=0.0000 | Acc=0.3400
 | AvgThinkTokens=149.44 | AvgBudget=149.44 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=32.61
 Recent budgets: [146, 137, 191, 171, 192, 157, 114, 116, 142, 147]
 [Epoch 5/5] Step 60/200 | AvgReward=0.6537 | NormRewardMean=-0.0000 | Acc=0.3500
 | AvgThinkTokens=147.10 | AvgBudget=147.10 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=31.57
 Recent budgets: [115, 117, 158, 131, 179, 107, 113, 149, 153, 132]
 [Epoch 5/5] Step 70/200 | AvgReward=0.6468 | NormRewardMean=0.0000 | Acc=0.3286
 | AvgThinkTokens=147.01 | AvgBudget=147.01 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=31.39
 Recent budgets: [173, 119, 168, 127, 175, 174, 164, 113, 87, 165]
 [Epoch 5/5] Step 80/200 | AvgReward=0.6291 | NormRewardMean=-0.0000 | Acc=0.3250
 | AvgThinkTokens=146.93 | AvgBudget=146.93 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=31.94
 Recent budgets: [192, 148, 120, 119, 202, 131, 193, 137, 132, 89]
 [Epoch 5/5] Step 90/200 | AvgReward=0.6261 | NormRewardMean=-0.0000 | Acc=0.3333
 | AvgThinkTokens=149.29 | AvgBudget=149.29 | MinBudget=85.00 | MaxBudget=211.00
 | StdBudget=31.65
 Recent budgets: [192, 168, 192, 180, 136, 138, 139, 184, 182, 171]
 [Epoch 5/5] Step 100/200 | AvgReward=0.6354 | NormRewardMean=-0.0000 |
 Acc=0.3500 | AvgThinkTokens=147.71 | AvgBudget=147.71 | MinBudget=85.00 |
 MaxBudget=211.00 | StdBudget=31.99
 Recent budgets: [85, 104, 144, 156, 102, 117, 172, 183, 115, 157]
 [Epoch 5/5] Step 110/200 | AvgReward=0.6443 | NormRewardMean=0.0000 | Acc=0.3636
 | AvgThinkTokens=147.35 | AvgBudget=147.35 | MinBudget=70.00 | MaxBudget=213.00
 | StdBudget=34.12
 Recent budgets: [182, 82, 70, 212, 148, 154, 113, 87, 177, 213]
 [Epoch 5/5] Step 120/200 | AvgReward=0.6353 | NormRewardMean=0.0000 | Acc=0.3583
 | AvgThinkTokens=145.88 | AvgBudget=145.88 | MinBudget=70.00 | MaxBudget=213.00
 | StdBudget=34.17
 Recent budgets: [118, 142, 120, 144, 103, 99, 195, 163, 90, 123]
 [Epoch 5/5] Step 130/200 | AvgReward=0.6301 | NormRewardMean=-0.0000 |
 Acc=0.3538 | AvgThinkTokens=145.54 | AvgBudget=145.54 | MinBudget=70.00 |
 MaxBudget=213.00 | StdBudget=33.57
 Recent budgets: [120, 147, 200, 120, 125, 139, 146, 162, 146, 109]
 [Epoch 5/5] Step 140/200 | AvgReward=0.6286 | NormRewardMean=0.0000 | Acc=0.3571
 | AvgThinkTokens=146.49 | AvgBudget=146.49 | MinBudget=70.00 | MaxBudget=213.00
 | StdBudget=34.08
 Recent budgets: [207, 162, 110, 185, 173, 163, 93, 179, 204, 113]
 [Epoch 5/5] Step 150/200 | AvgReward=0.6213 | NormRewardMean=0.0000 | Acc=0.3533
 | AvgThinkTokens=146.90 | AvgBudget=146.90 | MinBudget=70.00 | MaxBudget=213.00
 | StdBudget=33.86
 Recent budgets: [163, 152, 93, 172, 156, 101, 160, 189, 158, 182]
 [Epoch 5/5] Step 160/200 | AvgReward=0.6261 | NormRewardMean=-0.0000 |

```

Acc=0.3625 | AvgThinkTokens=145.91 | AvgBudget=145.91 | MinBudget=70.00 |
MaxBudget=213.00 | StdBudget=34.15
Recent budgets: [156, 84, 136, 196, 94, 153, 170, 109, 100, 112]
[Epoch 5/5] Step 170/200 | AvgReward=0.6346 | NormRewardMean=-0.0000 |
Acc=0.3824 | AvgThinkTokens=146.79 | AvgBudget=146.79 | MinBudget=70.00 |
MaxBudget=213.00 | StdBudget=34.26
Recent budgets: [165, 161, 195, 198, 182, 104, 172, 173, 95, 164]
[Epoch 5/5] Step 180/200 | AvgReward=0.6271 | NormRewardMean=0.0000 | Acc=0.3778
| AvgThinkTokens=146.62 | AvgBudget=146.62 | MinBudget=70.00 | MaxBudget=213.00
| StdBudget=34.23
Recent budgets: [106, 142, 128, 162, 199, 180, 111, 88, 157, 165]
[Epoch 5/5] Step 190/200 | AvgReward=0.6183 | NormRewardMean=0.0000 | Acc=0.3684
| AvgThinkTokens=146.82 | AvgBudget=146.82 | MinBudget=70.00 | MaxBudget=213.00
| StdBudget=34.11
Recent budgets: [160, 184, 152, 141, 122, 94, 145, 200, 119, 186]
[Epoch 5/5] Step 200/200 | AvgReward=0.6181 | NormRewardMean=-0.0000 |
Acc=0.3750 | AvgThinkTokens=147.38 | AvgBudget=147.38 | MinBudget=70.00 |
MaxBudget=213.00 | StdBudget=34.01
Recent budgets: [187, 195, 157, 86, 180, 182, 142, 142, 146, 163]

```

```

=====
Epoch 5 summary
Train Accuracy      : 0.3750
Train Avg ThinkTokens : 147.38
Train Avg Reward    : 0.6181
Train Avg Budget    : 147.38
Train Min Budget    : 70.00
Train Max Budget    : 213.00
Train Std Budget    : 34.01
Test Accuracy       : 0.3000
Test Avg ThinkTokens : 141.88
Test Avg Reward     : 0.5600
Test Avg Budget     : 141.88
Test Min Budget     : 140.00
Test Max Budget     : 143.00
Test Std Budget     : 0.71
=====

```

Loading best saved policy for final evaluation...

Final best policy evaluation on test set:

```

{
  "accuracy": 0.3,
  "avg_thinking_tokens": 141.88,
  "avg_reward": 0.5599598844609076,
  "avg_budget": 141.88,

```

```
"min_budget": 140.0,  
"max_budget": 143.0,  
"std_budget": 0.7110555533852472  
}
```

```
Saved final policy to two_stage_continuous_budget_policy_v5_final.pt  
Saved final value net to two_stage_value_net_v5_final.pt
```

```
[ ]: AI Disclouser: I have taken the help of ChatGPT while writing the code
```